Shuhong Ye

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https://github.com/kenn-san

EDUCATION

University of Nottingham Ningbo China Ningbo, China
BSc Hons Computer Science with Artificial Intelligence *Nov 2018 - Jul 2023*
Overall GPA: 4.0/4.0
Honors and Awards: Head's scholarship (2019)
Courses: Machine Learning (COMP3055 UNNC): **A+**, Computer Graphics (COMP3069 UNNC): **A+**

EXPERIENCE

Medical Image AI R&D Engineer Hefei, China
Hefei Qianshou Medical Technology Co., Ltd. *2024.05 - 2025.01*

- Focused on the R&D of an AI-assisted dermatological diagnosis model. Conducted literature reviews (e.g., ISIC Skin Image Analysis Workshop) to evaluate and select SOTA deep learning architectures for medical imaging. Trained, fine-tuned, and validated diagnostic models using PyTorch, successfully delivering a Proof of Concept (POC) with preliminary diagnostic capabilities.
- Led the collection, cleaning, and preprocessing of dermatological image data. Developed a data processing pipeline to address common medical data challenges, such as class imbalance and noise.

Video Question Answering Research Intern Ningbo, China
Intelligent Simulation and Digital Port Lab *Jan 2022 - Jun 2023*
Supervisor: *Prof. Jianfeng Ren*

- Defined a new set of problems called Online Open-ended Video Question Answering (O^2VQA). Designed and developed a Confidence-based Event-centric Online Video Question Answering (CEO-VQA) model using PyTorch. Constructed data processing pipelines in Python to construct and clean a large-scale VideoQA dataset consisting of 10,000 videos, establishing a robust benchmark. Work published in ICASSP 2023. **[1]**
- Proposed a visual-text attention mechanism to utilize the Contrastive Language-Image Pre-training (CLIP) to guide the crossmodal learning for VideoQA and address the insufficient linguistic supervision problem. Work accepted to ICIP 2023. **[2]**

Virtual Reality Laparoscopic Surgery Simulation Research Intern Ningbo, China
Intelligent Simulation and Digital Port Lab *Jul 2021 - Oct 2021*
Supervisor: *Prof. Ying Weng*

- Developed a laparoscopic surgery training application with Unity (C#) and Steam VR SDK. Implemented real-time soft tissue simulation for gallbladder and liver using Obi Physics packages.

Remote Sensing Image Processing Research Intern Ningbo, China
UNNC School of Geographical Sciences *May 2021 - Jul 2021*
Supervisor: *Prof. Meili Feng*

- Developed remote sensing image processing tools with Google Earth Engine API, Python and R for cloud removal and water remote sensing.

PUBLICATIONS

[1] Weikai Kong*, **Shuhong Ye***, Chenglin Yao, Jianfeng Ren. Confidence-based Event-centric Online Video Question Answering on a Newly Constructed ATBS Dataset, accepted to **IEEE International Conference on Acoustics, Speech and Signal Processing *(ICASSP '23)*. (*equal first authors)** **[pdf]** **[project]** **[code]** **[dataset]**

[2] **Shuhong Ye***, Weikai Kong*, Chenglin Yao, Jianfeng Ren, Xudong Jiang. Video Question Answering Using CLIP-Guided Visual-Text Attention, accepted to **IEEE International Conference on Image Processing *(ICIP '23)*. (*equal first authors)** **[pdf]** **[code]**

SELECTED PROJECT

AIGC Image Generation Workflow & Model Fine-Tuning *May 2023 - May 2026*
Independent Developer, Personal Project

- Deployed and maintained a ComfyUI instance on cloud, configuring underlying environments and optimizing VRAM usage for efficient AI image generation.
- Developed LoRA fine-tuning pipelines for foundational models (SDXL, FLUX, WAN series). Executed custom dataset annotation and optimized reusable ComfyUI workflows to achieve targeted, style-specific image generation.

A Framework for the Competition of Searching Algorithms *Nov 2021 - Apr 2022*
Tech Leader, Software Engineering Group Project, supervised by Prof. Jiawei Li

- Developed a framework using Java to evaluate the performance of algorithms on 4 optimization problems, multi-armed bandits, bin packing, portfolio selection and job shop scheduling. Led a 5-person team using Git/GitHub for version control and agile collaboration.
- Built a modular system with well-defined APIs, automating algorithm evaluation pipeline with clear visualization. Developed unit tests and integration tests to verify the robustness of 4 optimization algorithms across diverse boundary conditions.

SKILLS

Programming Skills: Python, C/C++, Java **Technologies:** PyTorch, OpenCV, Unity, LaTeX
Developer Tools: Git, GitHub, Linux (Ubuntu), VS Code, Docker **Testing & QA:** pytest, unittest, Selenium
Data & Databases: NumPy, SQL, Pandas
Languages: TOEFL 100 (Speaking: 22, Writing: 25), CET-6